

INDUSTRIAL TRAINING REPORT

Gene – Gene Function Prediction using RAG and Neo4j

Submitted in partial fulfilment of the
Requirements for the award of
Degree of Bachelor of Technology in Information Technology



Session: 2025-26

Submitted By

Name: Shristi Patel

University Roll No. : 23ESKIT160 **Semester/Branch:** 5ITC (G1)

SUBMITTED TO:

Department of Information Technology
SWAMI KESHVANAND INSTITUTE OF TECHNOLOGY, MANAGEMENT &
GRAMOTHAN, JAIPUR

DECLARATION

I hereby declare that the Industrial Training Report entitled "Gene – Gene Function Prediction using RAG and Neo4j" is an authentic record of my own work as requirements of 6-weeks Industrial Training during the period from 16th June 2025 to 30th July 2025 for the award of degree of B.Tech. (Information Technology), Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur, under the guidance of (Dr. Priyanka Yadav).

Shristi Patel
23ESKIT160

Date: _____

Certified that the above statement made by the student is correct to the best of our knowledge and belief

Examined by:

Dr. Priyanka Yadav

Dr. Vipin Jain
Head of Department

31 July 2025

CERTIFICATE OF COMPLETION

This is to certify that **Ms. Shristi Patel**, a student of Information Technology from Swami Keshvanand Institute of Technology, Jaipur, worked as a student intern under my supervision on the project *"Integrating Knowledge Graph with Generative Large Language Model-based Tools to make an AI-based Helper for Researchers in the field of Computational Biology"* in the Department of Computational Biology at Indraprastha Institute of Information Technology-Delhi (IIIT-Delhi).

She has worked here for 45 days from June 16, 2025, to July 30, 2025. During this period, she actively contributed to coding, data processing, and integrating AI-based solutions with knowledge graph structures. She explored and implemented new concepts in Natural Language Processing, Retrieval-Augmented Generation (RAG), and Python-based development, while also enhancing her understanding of computational biology. Her performance in terms of learning, applying new skills, and delivering results was excellent.

Sincerely



Dr Vibhor Kumar
Associate Professor
Department of Computational Biology
Indraprastha Institute of Information Technology, Delhi

ACKNOWLEDGEMENT

First and foremost, I wish to express my sincere thanks and gratitude to my esteemed Mentor, **“Dr. Vibhor Kumar”** who has contributed so much for successful completion of my Industrial Training by his thoughtful reviews and valuable guidance.

Next I would like to tender my sincere thanks to **“Dr. Vipin Jain ”** (Head of Information Technology) for his co-operation and encouragement. I am thankful to **Mr. Praveen Yadav**, Coordinator III year In-house internship 2025-26, for his support, cooperation and motivation provided during the internship for constant inspiration, presence and blessings.

Shristi Patel

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